

Week 7 Summary: Density Matrices, Channels & CPTP Maps

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1 Why State Vectors Are Not Enough

A pure state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ captures *quantum* uncertainty (measurement randomness), but it cannot represent *classical* uncertainty (our ignorance about which state was prepared).

Key example. The superposition $|+\rangle$ and the classical mixture “ $|0\rangle$ or $|1\rangle$ with probability $\frac{1}{2}$ each” give identical Z -measurement statistics but are distinguishable by an X measurement. No single state vector can describe the mixture — we need the **density matrix**.

2 The Density Matrix

Definition 2.1 (Density Matrix). For a system in pure state $|\psi_i\rangle$ with probability p_i ($p_i \geq 0$, $\sum_i p_i = 1$):

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|.$$

A single term gives a *pure* state; otherwise the state is *mixed*.

2.1 Three Defining Properties

A matrix $\hat{\rho}$ is a valid density matrix if and only if:

1. **Hermitian:** $\hat{\rho} = \hat{\rho}^\dagger$ (eigenvalues are real).
2. **Unit trace:** $\text{Tr} \hat{\rho} = 1$ (populations sum to 1).
3. **Positive semidefinite:** $\langle\phi|\hat{\rho}|\phi\rangle \geq 0 \quad \forall |\phi\rangle$ (all eigenvalues ≥ 0).

2.2 Three Key States

$$\hat{\rho}_{|0\rangle} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \hat{\rho}_{|+\rangle} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \hat{\rho}_{I/2} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Vocabulary: diagonal entries ρ_{jj} are *populations*; off-diagonal entries ρ_{jk} ($j \neq k$) are *coherences*. Noise typically destroys coherences while leaving populations roughly intact.

2.3 Purity

$$\gamma \equiv \text{Tr}(\hat{\rho}^2), \quad \frac{1}{d} \leq \text{Tr}(\hat{\rho}^2) \leq 1.$$

$\gamma = 1$ for pure states; $\gamma = 1/d$ for the maximally mixed state I/d .

3 Measurement & the Trace Formula

For any observable $\hat{O}$:

$$\langle\hat{O}\rangle = \text{Tr}(\hat{O}\hat{\rho}).$$

For a projective measurement $\{P_m\}$ with $\sum_m P_m = I$:

$$\text{Pr}(m) = \text{Tr}(P_m\hat{\rho}), \quad \hat{\rho} \xrightarrow{\text{outcome } m} \frac{P_m\hat{\rho}P_m}{\text{Tr}(P_m\hat{\rho})}.$$

4 The Bloch Sphere for Mixed States

Every single-qubit density matrix has a unique **Bloch representation**:

$$\hat{\rho} = \frac{1}{2}(I + r_x X + r_y Y + r_z Z) = \frac{1}{2} \begin{pmatrix} 1 + r_z & r_x - ir_y \\ r_x + ir_y & 1 - r_z \end{pmatrix}, \quad \vec{r} \in \mathbb{R}^3, \quad |\vec{r}| \leq 1.$$

State	$\vec{r}$	$\text{Tr} \hat{\rho}^2$	Location
$ 0\rangle$	$(0, 0, 1)$	1	North pole
$ +\rangle$	$(1, 0, 0)$	1	Equator, $+x$
Any pure	$ \vec{r} = 1$	1	Surface
Partially mixed	$ \vec{r} < 1$	$\frac{1}{2}(1 + \vec{r} ^2)$	Inside
$I/2$	$(0, 0, 0)$	$\frac{1}{2}$	Centre

Key insight: Pure states live on the surface; mixed states live inside. Every noise channel shrinks, squashes, or displaces the Bloch ball.

5 Composite Systems & the Partial Trace

For a two-qubit state $\hat{\rho}_{AB}$ on $\mathbb{C}^2 \otimes \mathbb{C}^2$, the **reduced state** of qubit A is obtained by tracing out B :

$$\hat{\rho}_A = \text{Tr}_B(\hat{\rho}_{AB}) = \sum_b (I_A \otimes \langle b|) \hat{\rho}_{AB} (I_A \otimes |b\rangle).$$

Bell state example: $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is globally pure ($\text{Tr} \hat{\rho}_{AB}^2 = 1$), but $\hat{\rho}_A = I/2$ is maximally mixed ($\text{Tr} \hat{\rho}_A^2 = \frac{1}{2}$).

This is decoherence: A qubit that interacts with its environment becomes entangled with it. Tracing out the environment makes the qubit appear mixed — coherences leak into correlations we cannot access.

6 Quantum Channels & the Kraus Representation

6.1 Closed vs. Open Systems

- **Closed:** $\hat{\rho} \rightarrow U \hat{\rho} U^\dagger$ (unitary, reversible, Bloch vector rigidly rotated).
- **Open:** System S interacts with environment E (initially in $|e_0\rangle$), then E is traced out.

6.2 Operator-Sum (Kraus) Form

Theorem 6.1 (Kraus Representation).

$$\hat{\rho}' = \mathcal{E}(\hat{\rho}) = \sum_k K_k \hat{\rho} K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I.$$

The K_k are called **Kraus operators**; the completeness relation $\sum_k K_k^\dagger K_k = I$ guarantees trace preservation.

6.3 CPTP Properties

A map $\mathcal{E}$ is a legitimate physical process iff it is:

- Linear** in $\hat{\rho}$.
- Trace-preserving (TP):** $\text{Tr} \mathcal{E}(\hat{\rho}) = \text{Tr} \hat{\rho}$.
- Completely positive (CP):** $\mathcal{E} \otimes \mathcal{I}$ maps every positive operator to a positive operator.

7 The Four Standard Single-Qubit Channels

Channel	Kraus operators	Bloch action	Models
Bit flip	$\sqrt{1-p} I, \sqrt{p} X$	$(r_x, (1-2p)r_y, (1-2p)r_z)$	X errors
Phase flip	$\sqrt{1-p} I, \sqrt{p} Z$	$((1-2p)r_x, (1-2p)r_y, r_z)$	T_2 dephasing
Depolarising	$\sqrt{1-3p/4} I, \sqrt{p/4} \sigma_j$	$(1-p) \vec{r}$	Avg. gate error
Amplitude damping	$\text{diag}(1, \sqrt{1-\gamma}), \sqrt{\gamma} 0\rangle\langle 1 $	$(\sqrt{1-\gamma} r_x, \sqrt{1-\gamma} r_y, (1-\gamma)r_z + \gamma)$	T_1 relaxation

Key Relationships

- Phase flip with $p(t) = \frac{1}{2}(1 - e^{-t/T_2})$ gives coherence decay $\rho_{01}(t) = \rho_{01}(0) e^{-t/T_2}$ — the T_2 law.
- Amplitude damping alone gives coherence decay time $2T_1$. With additional dephasing:
 $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi} \Rightarrow T_2 \leq 2T_1$.
- The depolarising channel is the *only* channel treating all three Bloch axes identically — it is what randomised benchmarking measures.

8 Exercise Solutions (Summary)

8.1 Exercise 1: Bloch Vector

Given $\hat{\rho} = \frac{1}{2}1.60.3$
 0.30.4:

- Hermitian ✓, $\text{Tr} = 1$ ✓, eigenvalues > 0 ✓ $\Rightarrow$ valid density matrix.
- Bloch vector: $\vec{r} = (0.3, 0, 0.6)$, $|\vec{r}| = \sqrt{0.45} \approx 0.671$.
- Since $|\vec{r}| < 1$: **mixed state** (purity = 0.725).

8.2 Exercise 2: Partial Trace

- $|+\rangle \otimes |0\rangle$ (product): $\hat{\rho}_A = |+\rangle\langle+| \Rightarrow$ **pure** ($\text{Tr } \hat{\rho}_A^2 = 1$).
- $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ (entangled): $\hat{\rho}_A = I/2 \Rightarrow$ **maximally mixed** ($\text{Tr } \hat{\rho}_A^2 = \frac{1}{2}$).

8.3 Exercise 3: Kraus Completeness

$$\sum_{k=0}^3 K_k^\dagger K_k = \left(1 - \frac{3p}{4}\right)I + 3 \cdot \frac{p}{4}I = I. \quad \checkmark$$

8.4 Exercise 4: Composition of Phase-Flip Channels

$$p_{\text{eff}} = p_1 + p_2 - 2p_1p_2, \quad 1 - 2p_{\text{eff}} = (1 - 2p_1)(1 - 2p_2).$$

For small p : $p_{\text{eff}} \approx p_1 + p_2$ (coherence shrinkage factors multiply).

Key Takeaways

1. The density matrix $\hat{\rho}$ is the complete description of a quantum state when classical uncertainty is present.
2. Noise = entanglement with the environment + partial trace. Every channel is a partial trace in disguise.
3. The Bloch picture: pure states on the surface, mixed states inside. Channels shrink the ball; mitigation tries to undo the shrinkage.
4. The Kraus representation $\mathcal{E}(\hat{\rho}) = \sum_k K_k \hat{\rho} K_k^\dagger$ with $\sum_k K_k^\dagger K_k = I$ is the standard tool for describing noise.